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Quotation number: 3

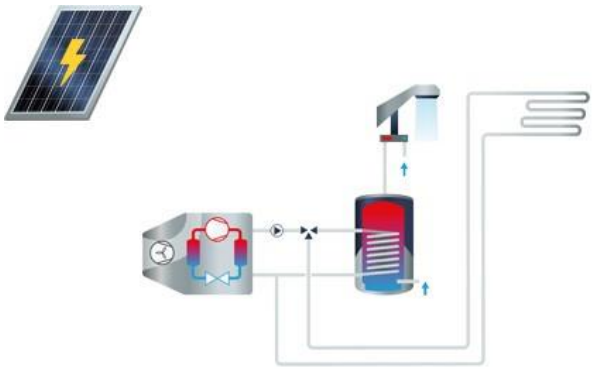
Optymalizacja systemu geotermalnego dla budynku jednorodzinnego (2+4) w miejscowości Rzeszów.

Rzeszów, ul. Konstantego Ildefonsa Gałczyńskiego 20



Projekt efektywnego systemu geotermalnego do ogrzewania i chłodzenia budynku jednorodzinnego (2+4) zmniejszający zużycie energii i redukujący emisję CO2 dzięki efektywnemu wykorzystaniu energii Ziemi. Projekt obejmuje analizę potencjału geotermalnego dla miejscowości Rzeszów, dobór odpowiednich wymienników ciepła, symulację wydajności systemu oraz analizę ekonomiczną i oszczędności energetycznych.

System: HP system with space heating and DHW



Results of the annual simulation

Utilized energy / Year

Space Heating:	11 382 kWh
Domestic hot water:	4 773 kWh

Energy generated / Year

Supplied by heat pump:	18 999 kWh

Seasonal Heat Performance Factors (from simulation)

SPF of heat pump:	3,0
SPF of heat pump system:	3,0

Power consumption / Year

Heat pump:	6 437 kWh
Auxiliary energy:	0 kWh

Losses / Year

Tank losses:	240 kWh
DHW recirculation loop:	2 607 kWh

Photovoltaic system

Generated PV Energy:	3 851 kWh
Of which own consumption:	1 622 kWh
Of which surplus:	1 435 kWh
Self-consumption share:	42,1 %
Total electr. energy consumed:	8 341 kWh
Of which covered by PV:	1 622 kWh
Of which purchased externally:	6 719 kWh
Degree of self-sufficiency:	19,4 %

Electrical energy consumed Heat pump:	6 437 kWh
Of which covered by PV:	491 kWh
Of which purchased externally:	5 945 kWh
Degree of self-sufficiency:	7,6 %

Electrical energy consumed Load profile:	1 905 kWh
Of which covered by PV:	1 131 kWh
Of which purchased externally:	774 kWh
Degree of self-sufficiency:	59,4 %

Primary energy savings and CO2 prevention

Primary energy savings:	17 761 kWh
CO2 emissions avoided:	4 263 kg

Site data

Climate data

Climate data record:	Rzeszow
Annual sum of global irradiation:	1 094 kWh/m²
Longitude:	-22,03 ° E
Latitude:	50,00 ° N

Domestic Hot Water

Average daily usage:	282 l
Target temperature:	50 °C
Cold water temperature in February:	6,5 °C
Cold water temperature in August:	12,5 °C
Circulation:	present
One-way length of piping system:	25 m
Temperature spread supply/return:	5,0 K
Specific losses:	0,30 W/(m*K)

Space Heating

Heated useable area:	130 m²
Indoor temperature:	21,0 °C
Heating limit temperature:	12 °C
Low temperature loop proportion:	100,0 %
Feed/Return of low temperature loop:	32,0/25,0 °C
Requirement type:	Heating output (yearly total)
Heating output (yearly total):	11 423 kWh
Resulting heating load:	7,1 kW

Non commercial

System components

Heat pump

Manufacturer:	Viessmann Werke GmbH & Co KG
Model:	AWO-M-AC (AF) 101.B06
Construction type:	Air/Water
Nominal heating power:	7,5 kW
Electr. nominal output:	1,8 kW
Mode of operation:	monovalent

Off-periods

No off-periods defined.

Ventilator

Nominal output:	0,000 kW
Flow rate:	2 329 m³/h

Single coil indirect hot water tank

Manufacturer:	Viessmann Werke GmbH & Co KG
Model:	Vitocell 300-V (300 l)
Volume:	300 l
Tank losses:	1,12 kWh/d

Photovoltaic system

Gross area:	32,6 m²
Total power:	7,2 kW
Inclination (tilt angle):	37 °
Orientation (north=0°, east=90°):	0 °
Azimuth (south=0°, west=90°):	-180 °

Battery capacity:	18 kWh
Use battery for heat pump:	No

Load profiles

1-person household	1 905 kWh

Financial Analysis

General parameters

Life span:	25 Years
Interest on capital:	10,00 %
Cost escalation rate:	2,00 %
Auxiliary electricity price (budget price):	0,300 zł/kWh

Tariffs

Basic monthly price:	20,00 zł
Electricity price:	0,200 zł/kWh

No HT/LT-tariffs defined.

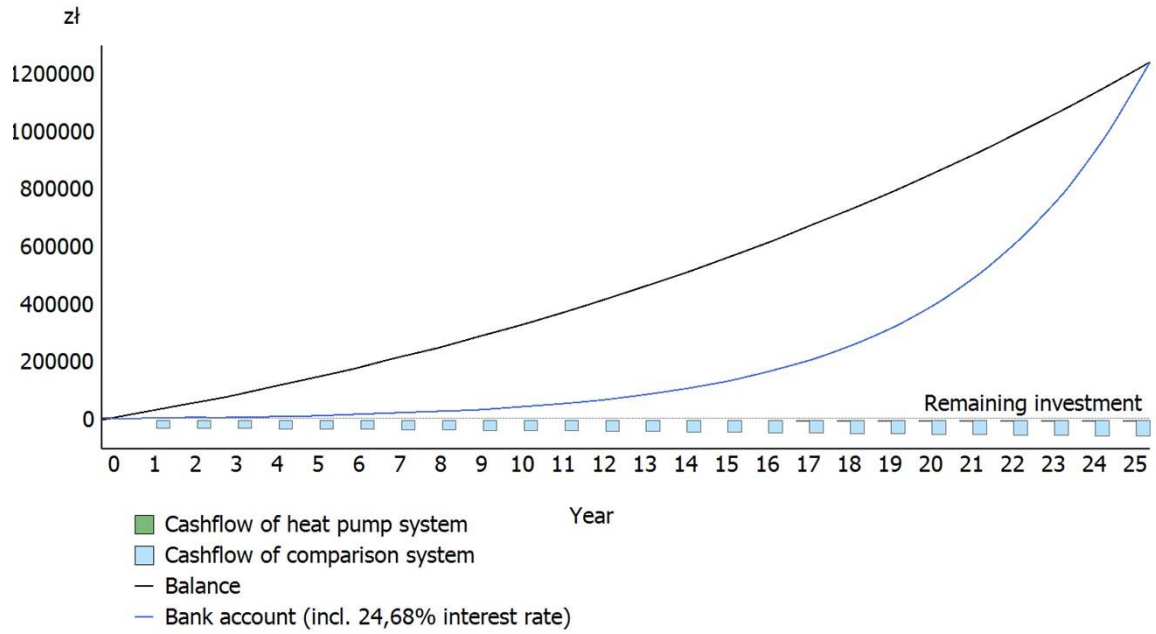
Financing

	Heat pump	Comparison system
Cost escalation rate of energy:	3,00 %	3,00 %
Energy price:	see Tariffs	1,100 zł/kWh
Investments (total):	10 000,00 zł	5 000,00 zł
Subsidies:	0,00 zł	0,00 zł
Maintenance costs:	200,00 zł	200,00 zł

Results

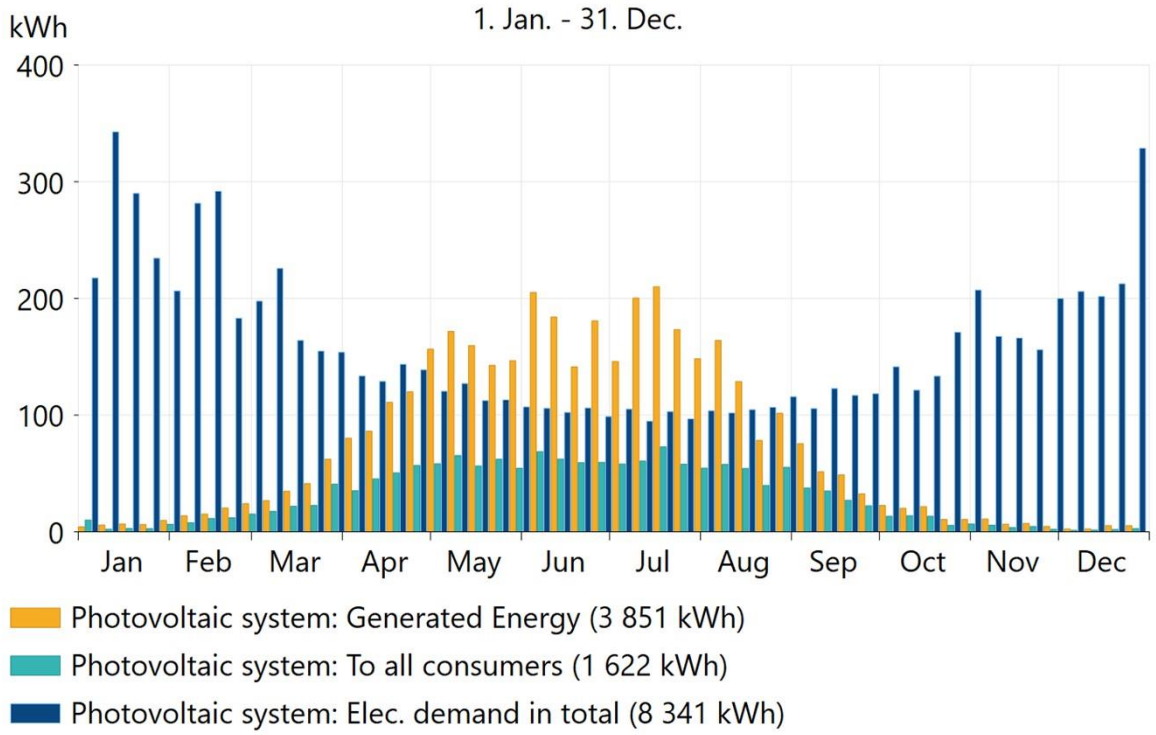
Net present value:		293 287,64 zł
Remaining investment:		5 000,00 zł
Return (MIRR):		24,68 %
Surplus:		1 236 461,71 zł
Investment annuity:	1 101,68 zł	550,84 zł
Operating costs annuity:	1 634,51 zł	34 496,28 zł
Maintanance costs annuity:	233,72 zł	233,72 zł
Heating price:	0,156 zł/kWh	1,857 zł/kWh

Cashflows and Balance

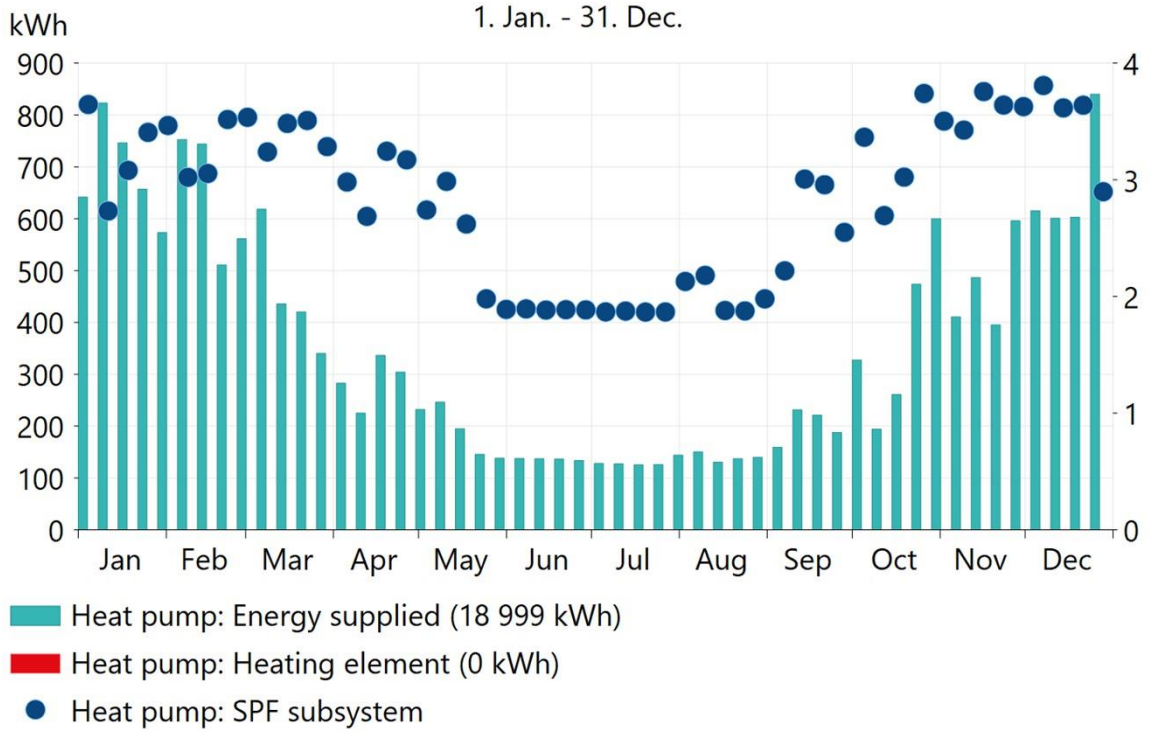

zł
1200000
1000000
800000
600000
400000
200000
0
Remaining investment
0 1 2 3 4 5 6 7 8 9 10 11 12 13 14 15 16 17 18 19 20 21 22 23 24 25
Year
Cashflow of heat pump system
Cashflow of comparison system
Balance
Bank account (incl. 24,68% interest rate)

Diagrams

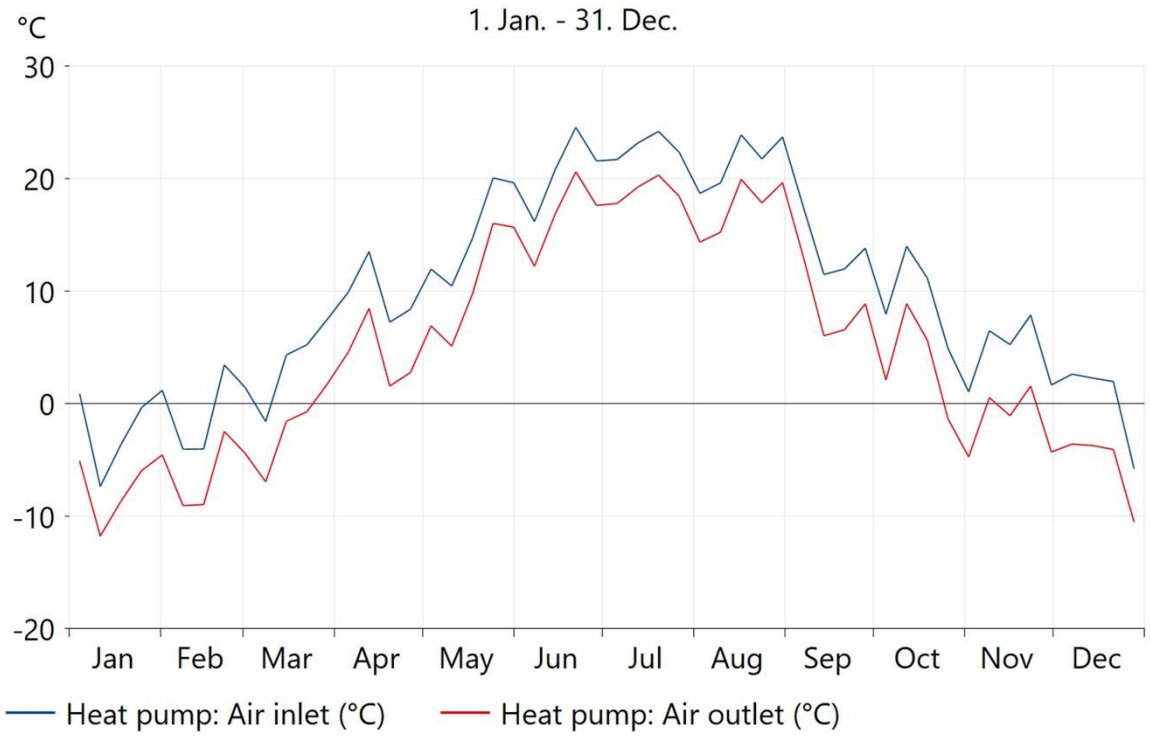
Heat pump with photovoltaics



Heat pump (energies and SPF)



Heat source temperatures (air)


°C
1. Jan. - 31. Dec.
30
20
10
0
-10
-20
Jan
Feb
Mar
Apr
May
Jun
Jul
Aug
Sep
Oct
Nov
Dec
Heat pump: Air inlet (°C)
Heat pump: Air outlet (°C)

VDI 4650 Seasonal Performance Factor

Seasonal Performance Factor:	4,33

Boundary conditions

Heating limit temperature:	12 °C
Standard outdoor temperature:	-15,0 °C
Temp. difference at the condenser (operating):	7 K
Temp. difference at the condenser (check):	5 K

Heat pump

Heat source:	Ambient air
Medium:	Air/Water
Mode of operation:	monovalent
COP A-7/W35:	3,35
COP A2/W35:	4,27
COP A7/W35:	5,03

v

These calculations were carried out by GeoT*SOL 2025 (R1) - the simulation program for heat pumps. The results are determined by a mathematical model calculation with variable time steps of up to 6 minutes. Actual yields can deviate from these values due to fluctuations in climate, consumption, geothermal conditions and other factors.The system schema above does not represent and cannot replace a full technical drawing of the heat pump system.